

Quantitative Aptitude

Formula - Guide

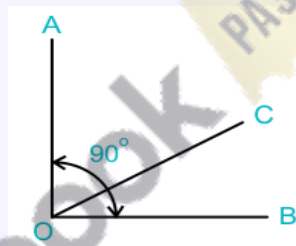
Geometry



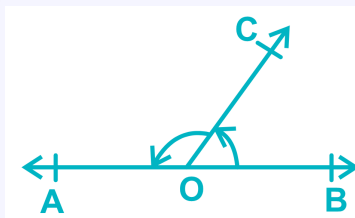
Geometry

Angles

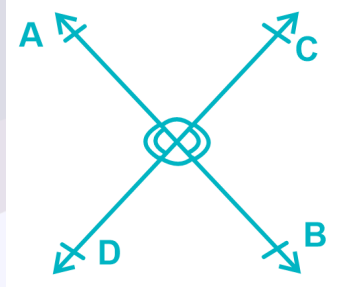
- **Acute angle** : $0^\circ < \theta < 90^\circ$
- **Right angle** : $\theta = 90^\circ$
- **Obtuse angle** : $90^\circ < \theta < 180^\circ$
- **Straight Angle** : $\theta = 180^\circ$
- **Reflex Angle** : $180^\circ < \theta < 360^\circ$
- **Complete Angle** : $\theta = 360^\circ$
- **Complementary Angles** : $\angle AOC + \angle COB = 90^\circ$



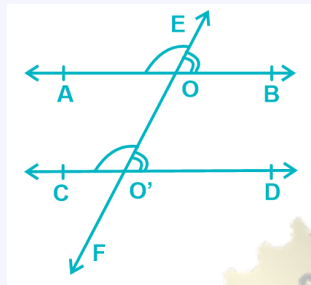
- **Supplementary Angles** : $\angle AOC + \angle COB = 180^\circ$



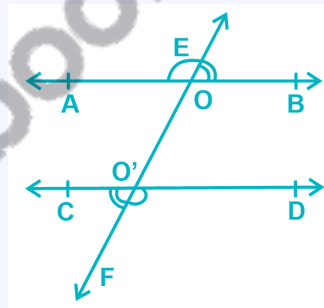
- **Vertically Opposite Angles** : $\angle AOC$ and $\angle BOD$ are vertically opposite angles. So, $\angle AOC = \angle BOD$ and $\angle BOC = \angle AOD$.



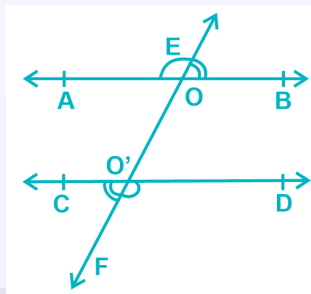
- **Corresponding Angles :** ($\angle BOE$, $\angle DO'O$) and ($\angle AOE$, $\angle CO'O$) are pairs of corresponding angles. So, $\angle BOE = \angle DO'O$ and $\angle AOE = \angle CO'O$.



- **Alternate Interior Angles :** ($\angle BOO'$, $\angle CO'O$) and ($\angle AOO'$, $\angle DO'O$) are the pairs of alternate interior angles. So, $\angle BOO' = \angle CO'O$ and $\angle AOO' = \angle DO'O$

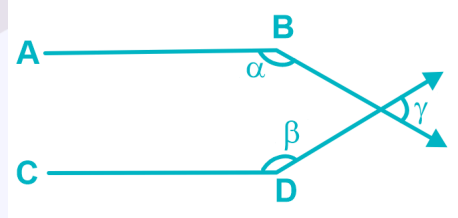


- **Alternate Exterior Angles :** ($\angle BOE$, $\angle CO'F$) and ($\angle AOE$, $\angle DO'F$) are the pairs of alternate exterior angles. So, $\angle BOE = \angle CO'F$ and $\angle AOE = \angle DO'F$

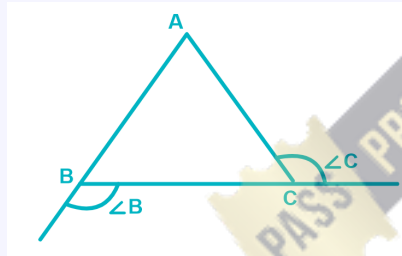


Some important results

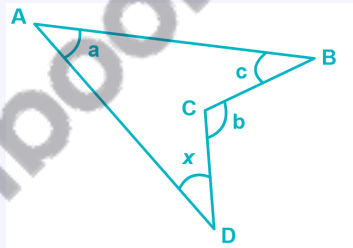
- If $AB \parallel CD$, then the value of $\alpha + \beta + \gamma = 360^\circ$



- The sum of the exterior angles at B and C is equal to $180^\circ + \angle A$.



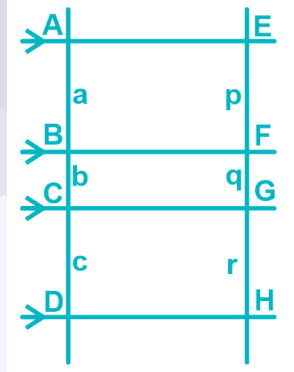
- Scissor Figure :** $b = a + c + x$



- When two transverse lines intersect four parallel lines :

$$a : b : c = p : q : r$$

$$a / (a+b+c) = p / (p + q + r)$$

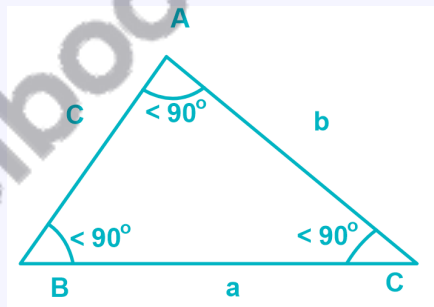


Triangle

Acute Angle Triangle

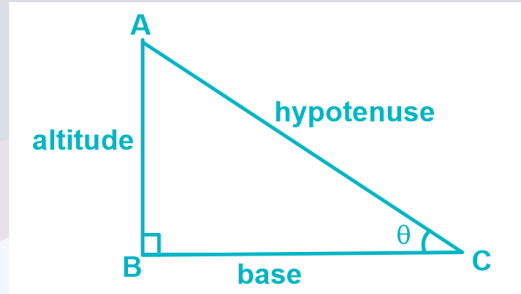
- All three angles $< 90^\circ$
- The sum of the squares of any two sides of the triangle is greater than the square of the third side.

$$AB^2 + BC^2 > AC^2, BC^2 + AC^2 > AB^2 \text{ and } AC^2 + AB^2 > BC^2$$



Right Angle Triangle

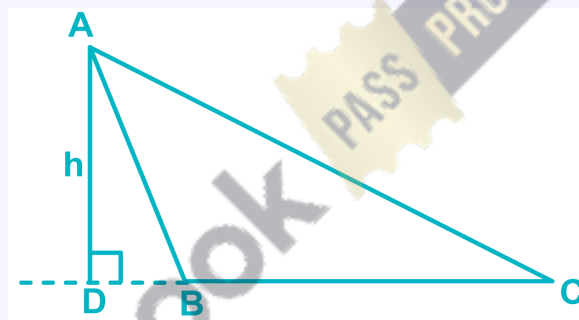
- One angles = 90°
- By Pythagoras theorem, $AB^2 + BC^2 = AC^2$



Obtuse Angle Triangle

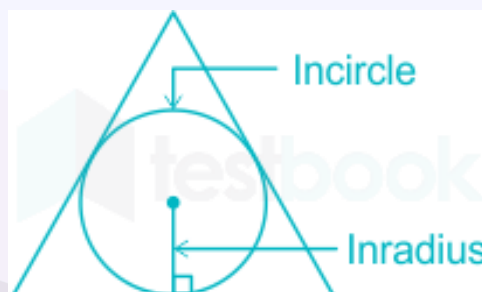
- One angles $> 90^\circ$
- The sum of the square of two smaller sides is less than the square of the third side (the largest side).

$$AB^2 + BC^2 < AC^2$$

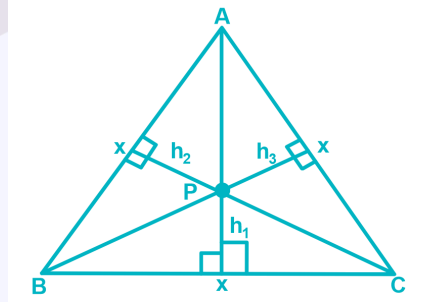


Equilateral Triangle

- Perimeter = $3a$
- Height of equilateral triangle = $\frac{\sqrt{3}}{2} a$
- Area of equilateral triangle = $\frac{\sqrt{3}}{4} a^2$
- Radius of the incircle (r) of an equilateral triangle = $(\text{side} \times \sqrt{3}) / 6$

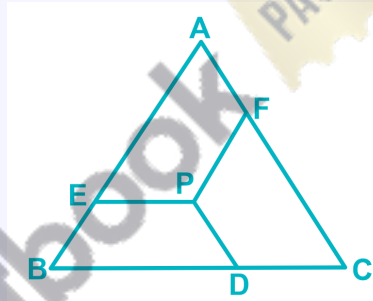


- The circumradius of an equilateral triangle is $= \frac{a}{\sqrt{3}}$
- If P is a point inside an equilateral triangle ABC, the sum of altitudes from the point P to the sides AB, BC and AC equal to the median of $\triangle ABC$.



$$h_1 + h_2 + h_3 = \frac{\sqrt{3}}{2} a = \text{Height of equilateral triangle.}$$

- If P is a point inside the triangle and $PF \parallel AB$, $PD \parallel AC$, $PE \parallel CB$.
then, $PD + PE + PF = AB = BC = CA$.



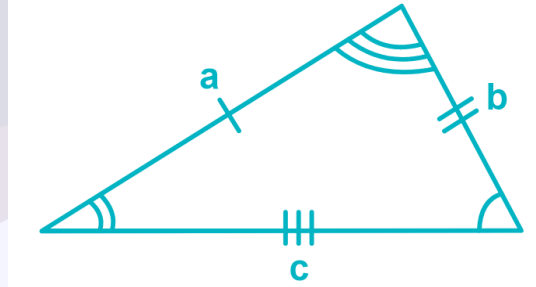
Isosceles Triangle

- Perimeter $= 2a + b$
- Area of triangle ABC $= \frac{b}{4} \sqrt{4a^2 - b^2} = \frac{1}{2} a^2 \sin \alpha$

Scalene Triangle

- Perimeter $= a + b + c$
- Semi perimeter $(s) = \frac{a+b+c}{2}$
- Area $= \sqrt{s(s-a)(s-b)(s-c)}$

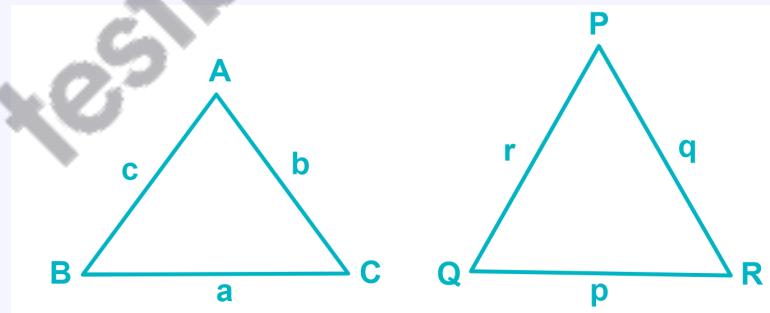
{ This is also known as Heron's Formula }



Similarity of Triangle

$\triangle ABC$ and $\triangle PQR$ are similar triangles:

- $\frac{a}{p} = \frac{b}{q} = \frac{c}{r}$
- $\angle A = \angle P, \angle B = \angle Q, \angle C = \angle R$
- Ratio of area = (Ratio of corresponding sides)²
- Ratio of perimeters of $\triangle ABC$ to $\triangle PQR$ = Ratio of corresponding sides.



General Properties of Triangle

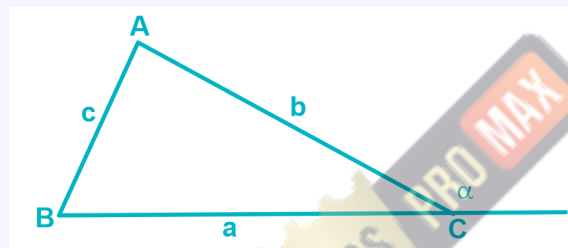
- **Property 1** - The sum of the three angles of a triangle is equal to 180° .
- **Property 2** - The sum of the two sides is greater than the third side.

$$a + b > c, a + c > b, b + c > a.$$

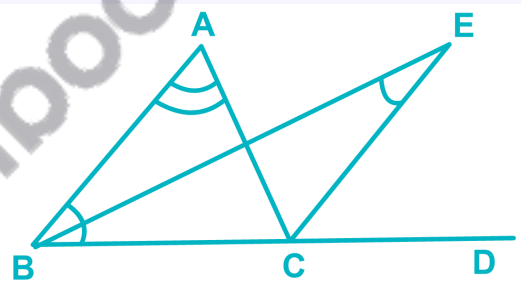
- **Property 3** - The difference between any two sides of a triangle is always smaller than the third side.

$$|a - b| < c, |b - c| < a, |c - a| < b$$

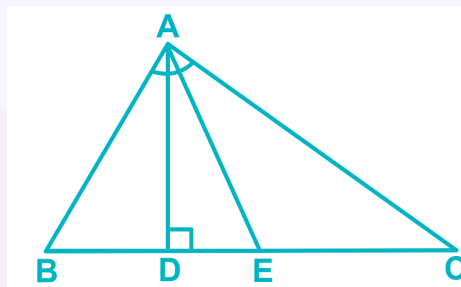
- **Property 4** - The exterior angle α is equal to the sum of two opposite interior angles A and B, i.e., $\alpha = \angle A + \angle B$.



- **Property 5** - In a triangle the angle between the internal bisector of a base angle and the external bisector of the other base angle is half of the remaining vertex angle. According to this property, $\angle BEC = \angle A / 2$



- **Property 6** - The angle between the perpendicular drawn by a vertex to the opposite side and the angle bisector of the vertex angle is half of the difference between the other two remaining vertex angles.

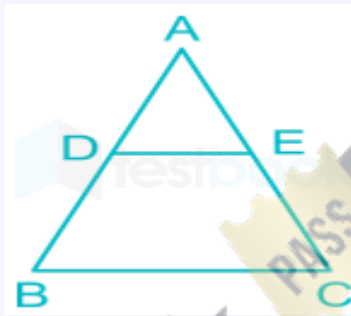


$AD \perp BC$ and AE is angle bisector of $\angle A$

Then, $\angle DAE = \left| \frac{B-C}{2} \right|$

- **Property 7** - Angles opposite to the equal sides of a triangle are equal. The angle opposite to the greater side is always greater than the angle opposite to the smaller side.

- **Property 8 - Basic Proportionality Theorem (Thales Theorem)**



- A line drawn parallel to one side of a triangle divides the other two sides in the same ratio.

i.e. If $DE \parallel BC$, then $\frac{AD}{DB} = \frac{AE}{EC}$

- If a line divides any two sides of a triangle in the same ratio, the line must be parallel to the third side.

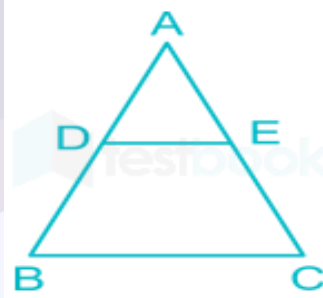
I.e. If $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$

- If $\triangle ADE \sim \triangle ABC$, then

$$\frac{Ar \triangle ADE}{Ar \triangle ABC} = \left(\frac{AD}{AB} \right)^2 = \left(\frac{AE}{AC} \right)^2 = \left(\frac{DE}{BC} \right)^2$$

- **Property 9 - Mid Point Theorem**

The line segment joining the mid-points of any two sides of a triangle is parallel to the third side and is half of the third side.



(i) If D and E are mid-points of AB and AC, respectively, then

$$DE \parallel BC \text{ and } DE = \frac{BC}{2}$$

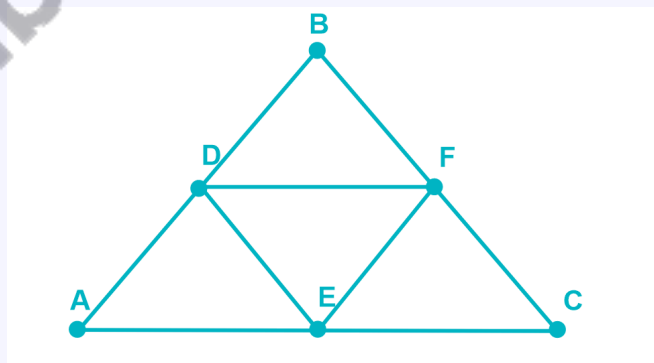
(ii) If $DE \parallel BC$ and $DE = \frac{BC}{2}$, then

D and E are the mid-points of AB and AC respectively.

- **Property 10** - By joining the mid points of the three sides of an equilateral triangle, we get a reverse equilateral triangle of one-fourth area of the original triangle. The sides of the new triangle will be half the original sides.

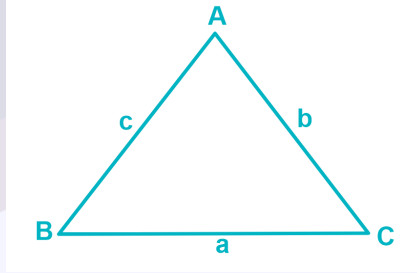
i.e. $\triangle ADE \cong \triangle FED \cong \triangle BDF \cong \triangle EFC$.

and $\text{Area}(\triangle ABC) = 4 \times \text{Area}(\triangle DEF)$



- **Property 11** -

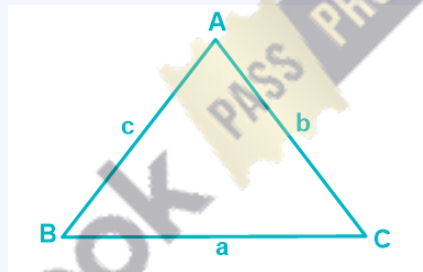
Sine Rule: Ratio of side and sine of opposite angle of a triangle is equal to double of circumradius (2R)



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = K \text{ (constant)}$$

• Property 12 -

Cosine Rule: If two sides and angle between sides are given, then we can find the opposite side by Cosine Rule.

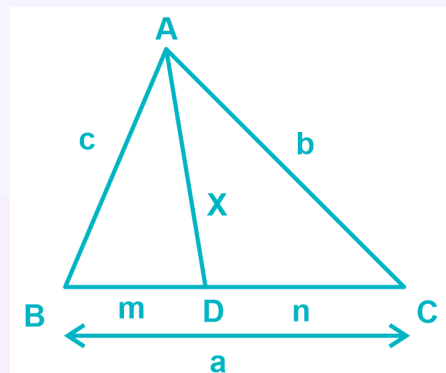


$$\cos A = (b^2 + c^2 - a^2) / 2bc$$

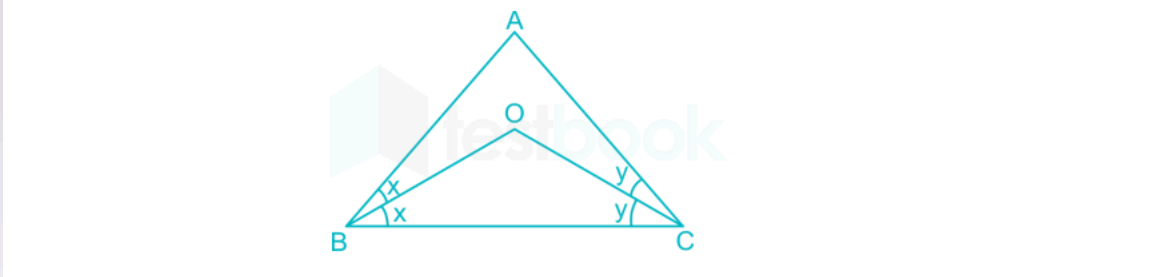
$$\cos B = (a^2 + c^2 - b^2) / 2ac$$

$$\cos C = (a^2 + b^2 - c^2) / 2ab$$

• Property 13 - Stewart's Theorem : $X^2 = \frac{b^2m + c^2n}{2} - mn$



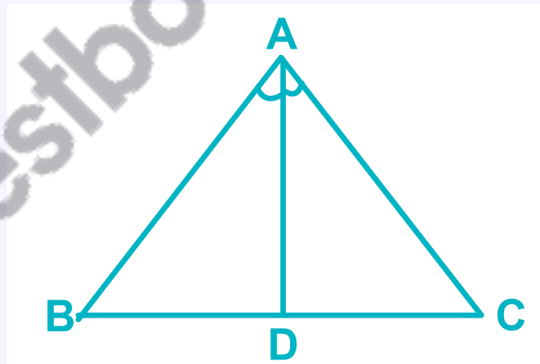
- **Property 14** - The internal bisectors of the angles B and C of a triangle ABC meet at O. Then, $\angle BOC = 90^\circ + (\angle A/2)$



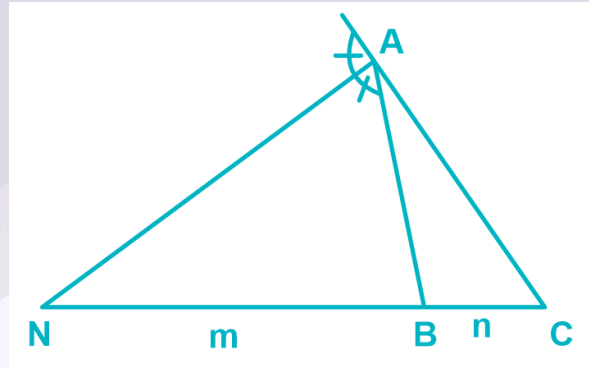
Centers of Triangle

Angle Bisector

A line that splits an angle into two equal parts. If line AD is such that $\angle BAD = \angle CAD$, then AD is the interior bisector of the angle A.

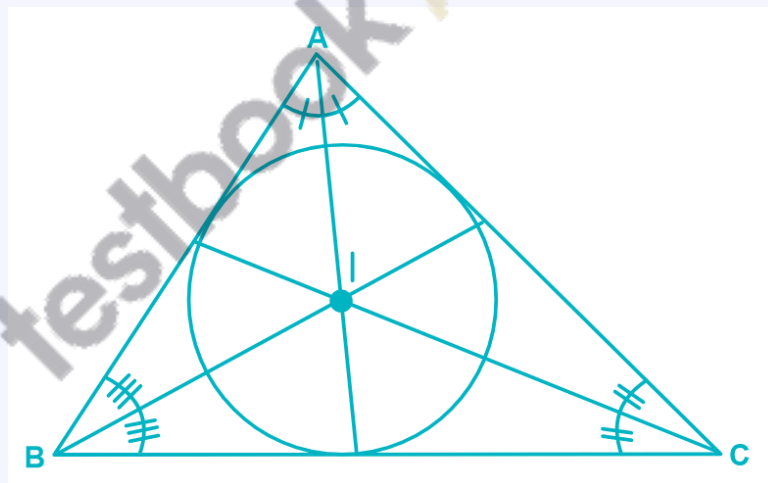


- (i) **Internal Angle Bisector** - The Internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the sides containing the angle, i.e. $\frac{BD}{CD} = \frac{AB}{AC}$
- (ii) **External Angle Bisector** - The external bisector of an angle of a triangle divides the opposite side externally in the ratio of the sides containing the angle. i.e. $\frac{BD}{CD} = \frac{AB}{AC}$



Incentre

The incentre of a triangle is the intersection point of all the three interior angle bisectors of the triangle.



- Using the angle sum property of a triangle, we can calculate the incentre of a triangle angle.

In the above figure,

$$\angle AIB = 180^\circ - (\angle A + \angle B)/2$$

Where I is the incentre of the given triangle.

- Incentre divides each angle bisector in the ratio of length of sum of two adjacent sides and opposite sides.

$$AI : ID = b + c : a$$

$$BI : IE = a + c : b$$

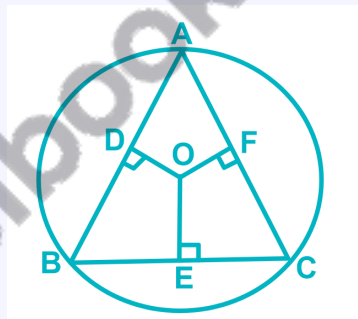
$$CI : IF = a + b : c$$

- The incenter of a triangle is equally inclined to all the sides of the triangle.

$$\text{Hence, Angle at incentre} = 90^\circ + \text{vertex angle}/2$$

Circumcentre

The point of intersection of the perpendicular bisectors of the sides of a triangle is called the circumcentre.



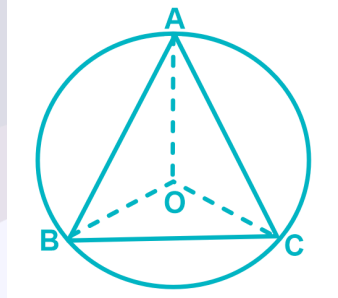
$$OD \perp AB \text{ and } AD = BD$$

$$OE \perp BC \text{ and } BE = EC$$

$$OF \perp AC \text{ and } CF = FA$$

- **In acute-angled triangle ABC,**

$$\angle BOC = 2\angle A, \quad \angle COA = 2\angle B \text{ \& \ } \angle AOB = 2\angle C$$



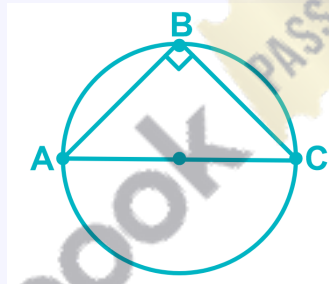
- **In a right-angled triangle, ABC**

Circumcenter of a right angled triangle is the midpoint of the hypotenuse of the right angled triangle.

$$\angle BOC = 2\angle A = 180^\circ$$

$$\angle COA = 2\angle B = 180^\circ - 2\angle C$$

$$\angle AOB = 2\angle C = 180^\circ - 2\angle B$$

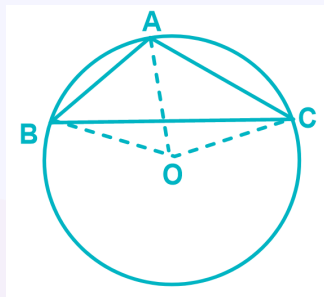


- **In an obtuse-angled triangle ABC,**

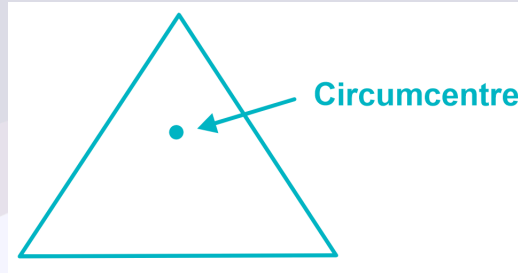
$$\angle BOC = 2(180^\circ - \angle A)$$

$$\angle COA = 2\angle B \text{ \& }$$

$$\angle BOA = 2\angle C$$



- **In an acute triangle,** the circumcentre is always inside the triangle.



- **In a right-angled triangle**, the circumcentre is mid-point of the hypotenuse.
- **In an obtuse-angled triangle**, the circumcentre is always outside the triangle and will be at the front of the angle which is obtuse.
- **Circumradius** : The radius of the circumcircle of a triangle is called the circumradius. It is denoted by R.

$$\text{The circumradius of any triangle ABC} = \frac{\text{Multiplication of all three sides}}{\text{Area of the triangle}} = \frac{abc}{4\Delta}$$

$$\text{The circumradius of a right-angled triangle} = \frac{\text{Hypotenuse}}{2} = \frac{h}{2}$$

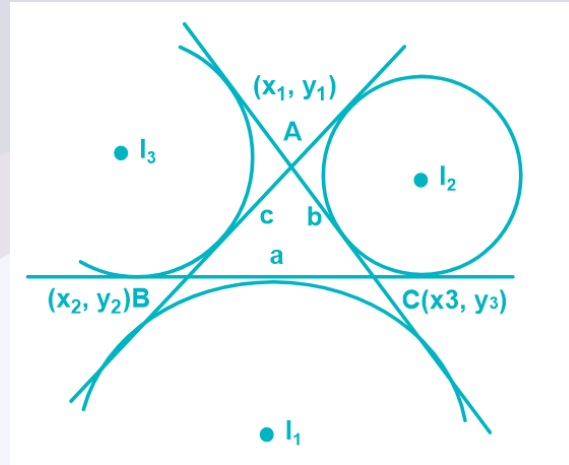
$$\text{The circumradius of an equilateral triangle} = \frac{\text{side}}{\sqrt{3}} = \frac{a}{\sqrt{3}}$$

Note : Distance between circum-center and incentre in any triangle is $= \sqrt{R^2 - 2Rr}$

Where, R = circum-radius and r = inradius

Excentre

Excentre of a triangle is the point of concurrency of bisectors of two exterior and third interior angle. Hence there are three excentres I_1 , I_2 and I_3 opposite to three vertices of a triangle.



- Ex- radii -**

$$r_a = \frac{\Delta}{s-a} , \quad r_b = \frac{\Delta}{s-b} , \quad r_c = \frac{\Delta}{s-c}$$

$$\text{Where, semi perimeter (s)} = \frac{a+b+c}{2}$$

$$\text{and } \Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

- Coordinates of centre of ex-circle opposite to vertex A are given as**

$$I_1(x, y) = (-ax_1+bx_2+cx_3/-a+b+c, -ay_1+by_2+cy_3/-a+b+c).$$

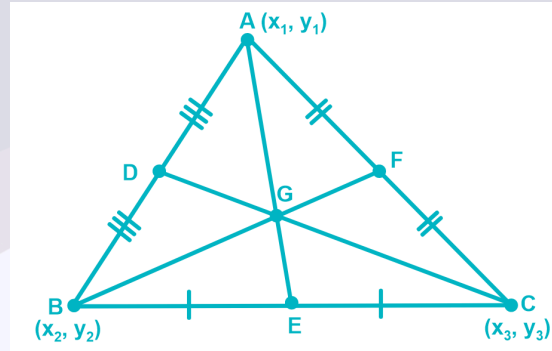
Similarly, coordinates of centre of $I_2(x, y)$ and $I_3(x, y)$ are

$$I_2(x, y) = (ax_1-bx_2+cx_3/a-b+c, ay_1-by_2+cy_3/a-b+c)$$

$$I_3(x, y) = (ax_1+bx_2-cx_3/a+b-c, ay_1+by_2-cy_3/a+b-c)$$

Centroid

The point of intersection of the medians of the triangle is called the centroid. It is denoted by G.



- Coordinates of centroid (G) = $(x_1 + x_2 + x_3 / 3, y_1 + y_2 + y_3 / 3)$
- If AD, BE and CF are medians of $\triangle ABC$ and G is the centroid,

$$\frac{AG}{GE} = \frac{BG}{GF} = \frac{CG}{GD} = \frac{2}{1}$$

- The sum of any two sides of a triangle is greater than twice the median drawn to the third side.

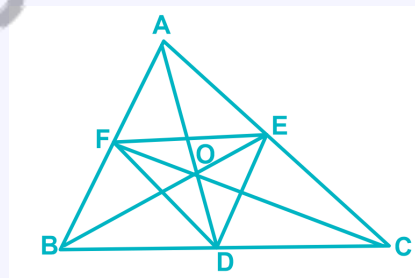
$$AB + AC > 2AE$$

$$AB + BC > 2BF$$

$$AC + BC > 2CD$$

- The area of the triangle formed by joining midpoints of two sides and centroid is $\frac{1}{12}$ of the area of the triangle.

$$\text{Ar } \triangle OFE = \text{Ar } \triangle OFD = \text{Ar } \triangle OED = \frac{1}{12} \text{Ar } \triangle ABC$$



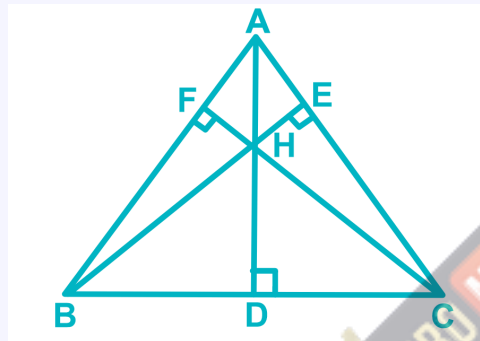
- The line segment joining the midpoints of two sides divides the line joining of vertex in between line to the centroid in the ratio in 3:1.
- If three medians of a triangle are given then,

$$\text{Area of } \triangle = \frac{4}{3} (\text{Area of triangle considering medians as side})$$

Orthocentre

The point of intersection of the altitudes of a triangle is called the orthocentre of the triangle. It is generally denoted by H.

All the three altitudes of a triangle pass through a point (orthocentre).

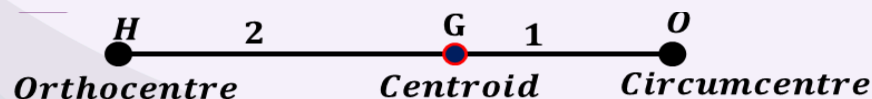


- In an acute-angled triangle the orthocentre is always inside the triangle.
- In a right-angled triangle the orthocentre is on the vertex of the triangle at which the triangle is right-angled.
- In an obtuse-angled triangle the orthocentre is always outside of the triangle and will be at the back of the angle which is obtuse.
- $\angle BHC = 180^\circ - \angle A$
 $\angle CHA = 180^\circ - \angle B$
 $\angle AHB = 180^\circ - \angle C$

It is true for all types of triangles.

Relation between Orthocentre, centroid and circumcentre –

In any triangle, orthocenter, centroid and circumcenter are collinear and centroid divides the line 2:1 as in figure –



Circle

A circle is a set of all points in a plane that are equidistant from a fixed point, called the center.

- Diameter = 2 (radius) = $2r$
- Circumference = $2\pi r$
- Area = πr^2

Semi-Circle

- Diameter = $2r$
- Perimeter = $r(2 + \pi)$
- Area = $\frac{\pi r^2}{2}$

Direct Common Tangent

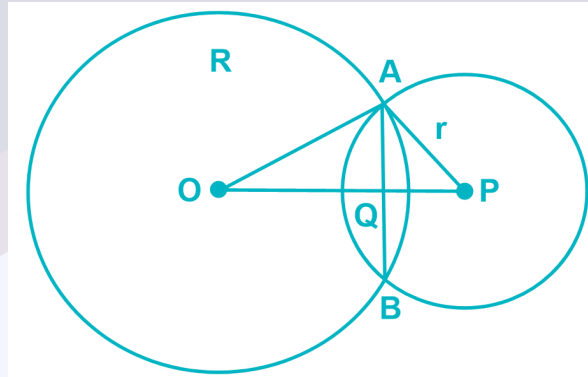
- *Length of Direct Common Tangent* = $\sqrt{d^2 - (r_1 - r_2)^2}$
- *Length of Transverse Common Tangent* = $\sqrt{d^2 - (r_1 + r_2)^2}$

Common Chord

- **Circles of radii R and r intersect and AB is the common chord, then**

$$\rightarrow AQ = BQ = \frac{AB}{2}$$

$$\rightarrow \text{Distance between the centres, } OP = \sqrt{R^2 - \left(\frac{AB}{2}\right)^2} + \sqrt{r^2 - \left(\frac{AB}{2}\right)^2}$$



- **When $\angle OAP = 90^\circ$** i.e. The tangent of a circle at point A or B passes through the centre of the other circle. Then,

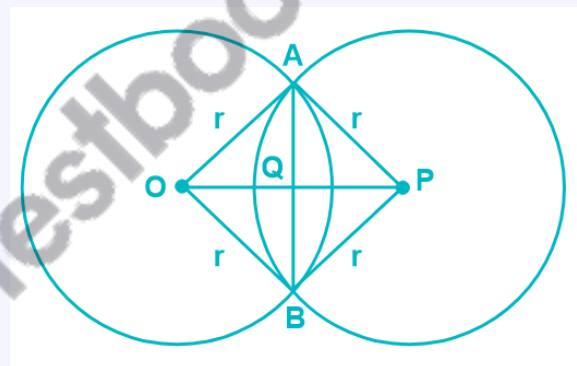
→ Distance between the circles, $OP = \sqrt{R^2 + r^2}$

→ The length of chord, $AB = \frac{2Rr}{OP}$

- **When radii of both the circles are equal, then $R = r$**

→ $OP = \sqrt{4R^2 - AB^2}$

→ $OQ = PQ = OP/2$

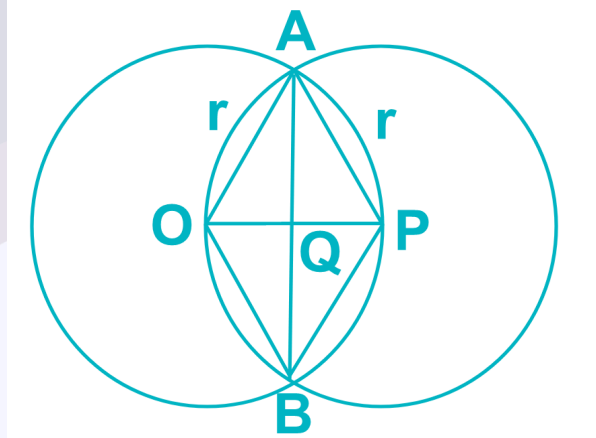


- **When radii of both the circles are equal and the both the circles pass through the centres of the other circle, i.e. $R = r$**

→ $\triangle OPA$ is an equilateral triangle.

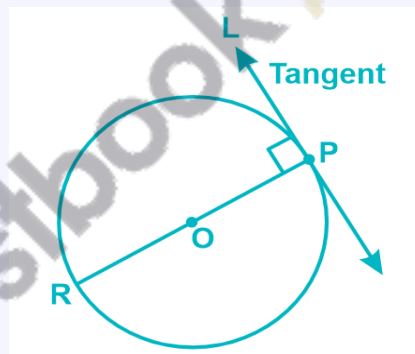
→ $OP = r$, $AB = \sqrt{3} r$

→ OBPA is a rhombus.

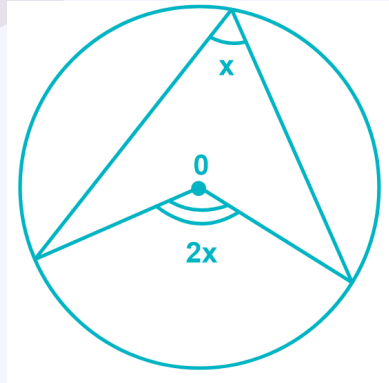


Important Theorems of Circle

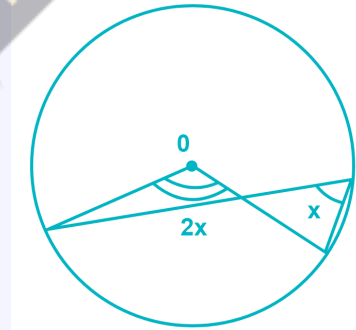
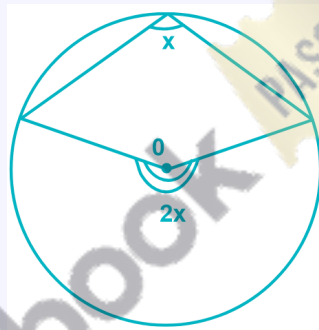
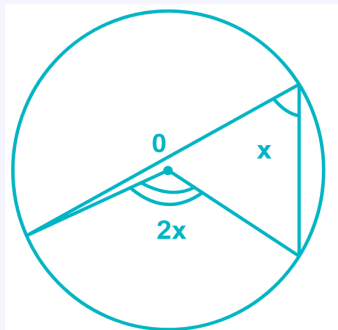
- Theorem 1** - When a radius of a circle meets a tangent at the point of tangency, the angle formed between the radius and the tangent is always 90° . ($\angle OPL = 90^\circ$)



- Theorem 2** - The angle subtended by an arc at the center of a circle is twice the angle subtended by the same arc at any point on the circumference of the circle. It is important that angle x is at the circumference of the circle.

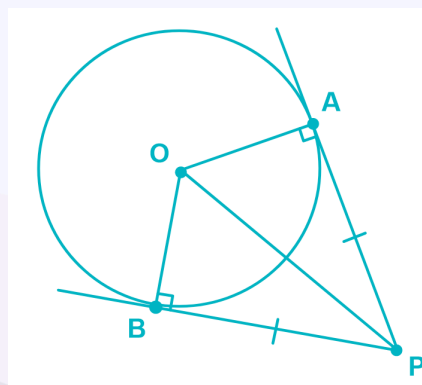


The following are other ways the theorem can be applied:

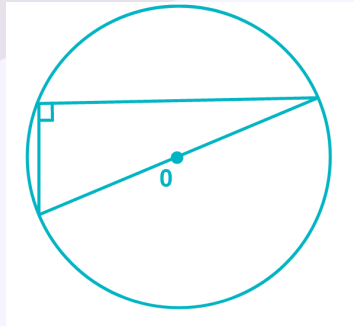


- Theorem 3** - Tangents drawn from an external point to a circle are equal in length from that point to the points of tangency on the circle.

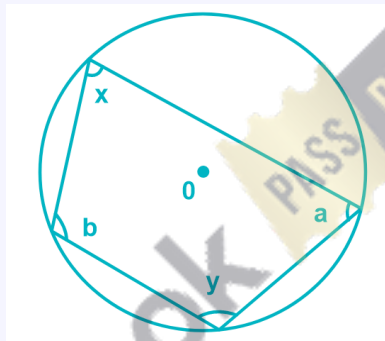
i.e. $PA = PB$



- **Theorem 4** - The angle subtended by a diameter of a circle at any point on the circumference of the circle (in a semicircle) is always a right angle (90°).



- **Theorem 5** - In a cyclic quadrilateral, opposite angles add up to 180° .

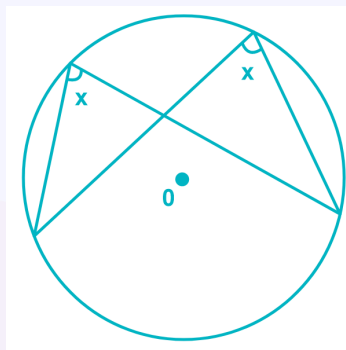


In the above diagram,

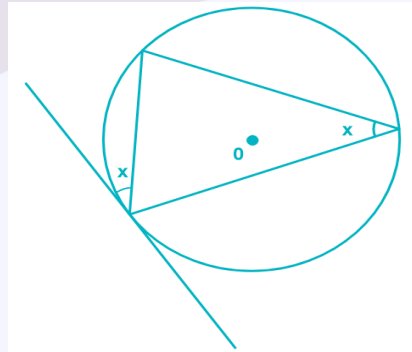
$$x + y = 180^\circ$$

$$a + b = 180^\circ$$

- **Theorem 6** - Angles subtended by the same arc at any points on the circumference of a circle (in the same segment) are equal.

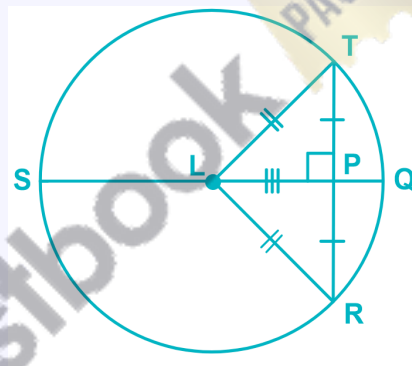


- **Theorem 7** - The angle formed between a tangent to a circle and a chord drawn from the point of tangency is equal to the angle subtended by that chord in the opposite (alternate) segment of the circle.



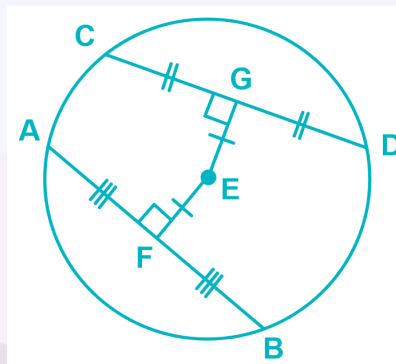
- **Theorem 8** - In a circle, a line drawn from the center of the circle perpendicular to a chord will bisect the chord, dividing it into two equal parts.

i.e. $\angle P = 90^\circ$ and $PT = PR$



- **Theorem 9** - In a circle, if two chords are congruent (equal in length), then they are equidistant from the center of the circle.

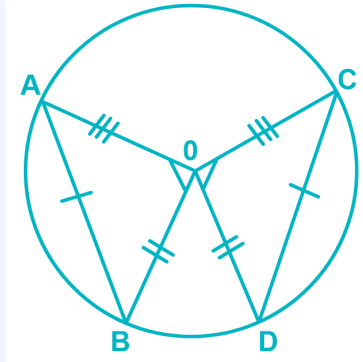
If $AB = CD$ Then $EG = EF$



- **Theorem 10** - Equal chords of a circle subtend equal angles at the centre.

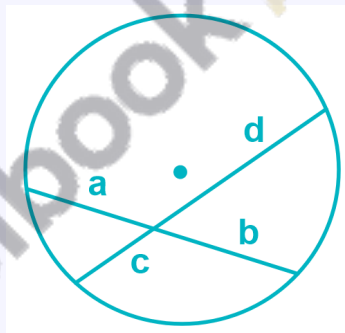
$\triangle AOB \cong \triangle COD$ (SSS) so, matching angles of congruent triangles are equal.

Angle AOB = Angle COD



- **Theorem 11** - If two chords intersect each other inside a circle, the product of the lengths of the segments of one chord is equal to the product of the lengths of the segments of the other chord.

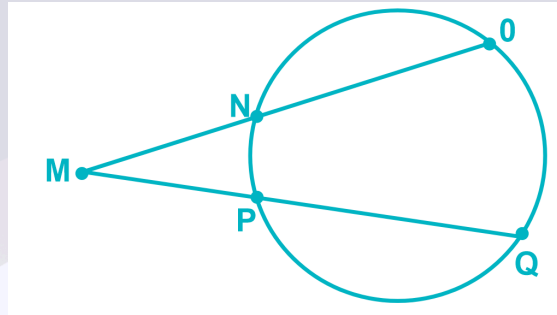
I.e. $a \times b = c \times d$



- **Theorem 12** - If two secant segments are drawn to a circle from an exterior point, then the product of the measures of one secant segment and its external secant segment is equal to the product of the measures of the other secant segment and its external secant segment.

In the circle, MO and MQ are secants that intersect at point M.

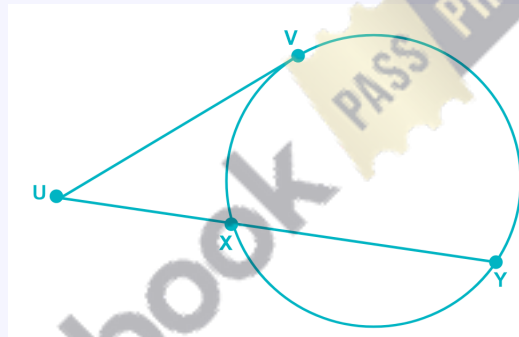
So, $MN \times MO = MP \times MQ$



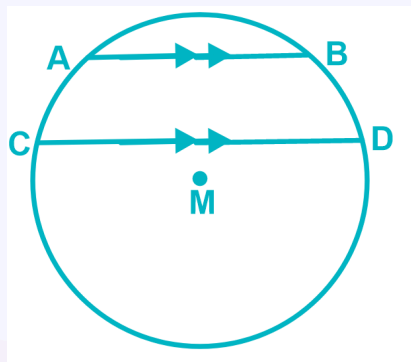
- **Theorem 13** - The product of the intercepts on a secant from an external point equals the square of the tangent from that point.

In the circle, UV is a tangent and UY is a secant, They intersect at point U.

$$\text{So, } UV^2 = UX \times UY$$

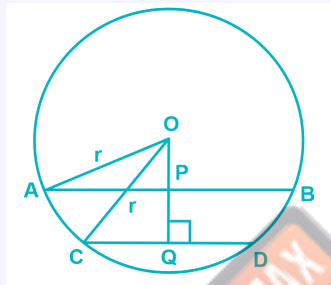
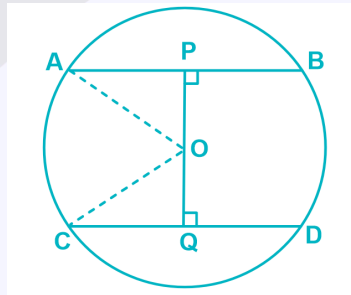


- **Theorem 14** - The measures of the arcs between parallel chords of a circle are equal. If $AB \parallel CD$ Then $\text{arc } AC = \text{arc } BD$



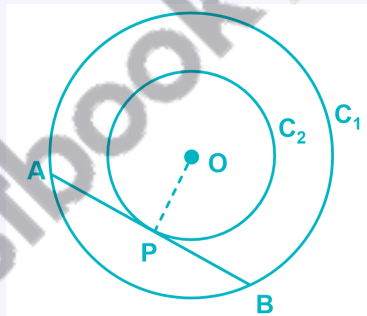
- Theorem 15** - In a circle distance between the chords is the sum of the distances from the center to the chords if the chords are on opposite sides of the center. If the chords are on the same side of the center, then the distance between them is the difference of the distances from the center to the chords.

Distance between Chords AB and CD = $OP + OQ$



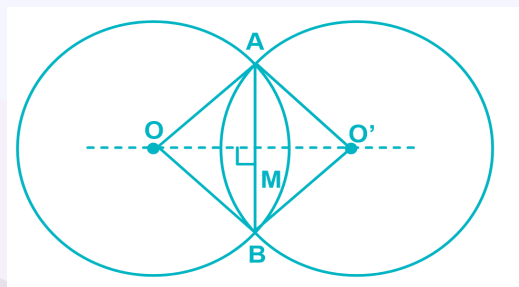
- Theorem 16** - In two concentric circles, if the chord of the outer circle touches the inner circle at one point, i.e. tangent to the inner circle, then the chord is bisected at the point of contact.

$\angle OPB = 90^\circ$ and $AP = BP$



- Theorem 17** - If two circles intersect at two points, then their centers lie on the perpendicular bisector of the common chord.

$\angle AMO = 90^\circ$ and $AM = BM$



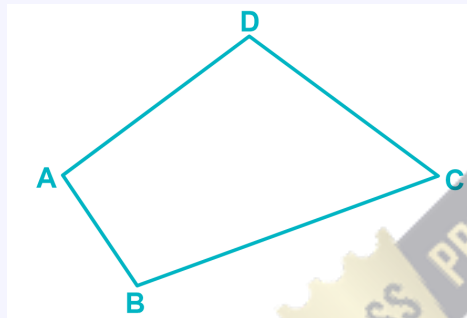
General Quadrilateral -

- Sum of interior angles of a quadrilateral = 360°

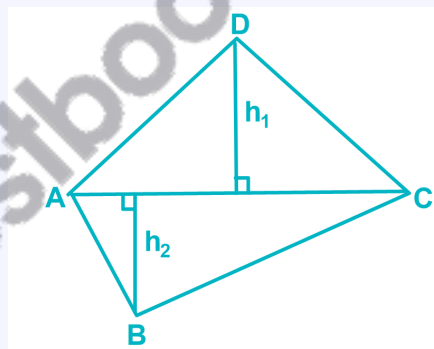
$$\text{i.e., } \angle A + \angle B + \angle C + \angle D = 360^\circ$$

- If diagonals of the quadrilateral intersect each other at 90° , then

$$AB^2 + CD^2 = BC^2 + AD^2$$



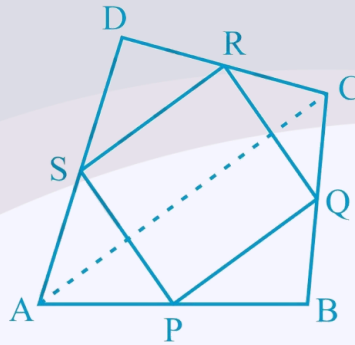
- Area of quadrilateral = $\frac{1}{2} \times \text{diagonal} \times (\text{sum of perpendicular dropped on it})$



- The quadrilateral formed by joining the midpoint of the adjacent side of the quadrilateral will be a parallelogram of half area.

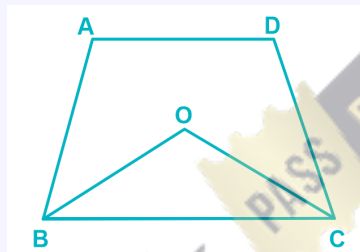
i.e. If the area of the quadrilateral is x ,

$$\text{then Area of the parallelogram PQRS} = \frac{x}{2}$$



- If BO and CO are the angle bisectors of angles $\angle B$ and $\angle C$, respectively, then

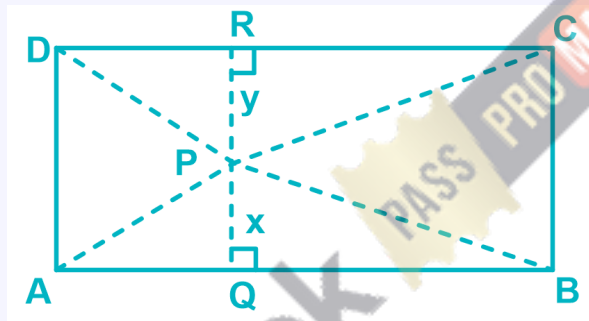
$$\angle BOC = \frac{1}{2} (\angle A + \angle D)$$



Rectangle

- A rectangle has four sides and four vertices. Hence, it's a quadrilateral.
- All the internal angles are equal to 90° .
- Opposite sides are equal and parallel.
- Diagonals are equal and bisect each other but not at right angles.
- The sum of any two adjacent angles is 180° .
- The sum of all the interior angles is 360° .
- Perimeter = $2(l + b)$; where, l = length and b = width
- Area (a) = $l \times b$
- All the rectangles of a given perimeter the square will have the maximum area.
- Bisectors of the four angles enclose a square.

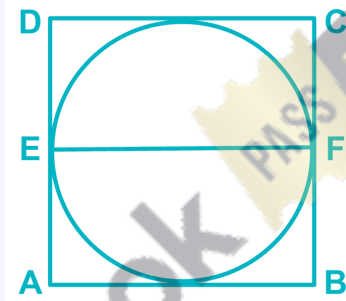
- A rectangle is a cyclic quadrilateral – that is, every one of its points would fit on the edge of a circle. The radius of that circle would be half of the rectangles diagonal in length.
- The diagonals of a rectangle are not perpendicular to each other unlike that of a square. A rectangle has two lines of symmetry. It has rotational symmetry around both vertical and horizontal axes.
- When the rectangle is inscribed in a circle, it will have the maximum area when it is a square.
- If P is any point inside the rectangle, then, $AP^2 + PC^2 = BP^2 + PD^2$



Square

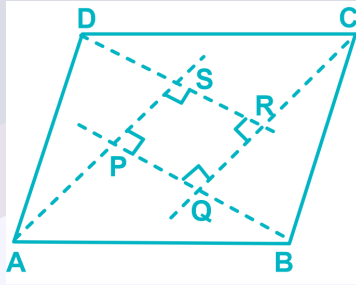
- All the sides of a square are equal.
- Each of the angles measures 90° .
- The sum of all the interior angles is 360° .
- The diagonals of a square are equal.
- The diagonals of a square bisect each other at right angles.
- Since the sides of a square are parallel, it is also called a parallelogram.
- The length of the diagonals in a square is greater than its sides.
- Perimeter = $4a$

- $\text{Area} = a^2 = (\text{diagonal})^2 / 2 = \frac{d^2}{2}$
- $\text{Diagonal} = a\sqrt{2}$
- Straight lines joining the mid points of adjacent sides of any square forms a square.
- If you rotate a square by any angle, each interior angle is still 90° . The rotation doesn't change the angles; it only changes the orientation of the square.
- The side of a circumscribed square is equal to the diameter of the inscribed circle.



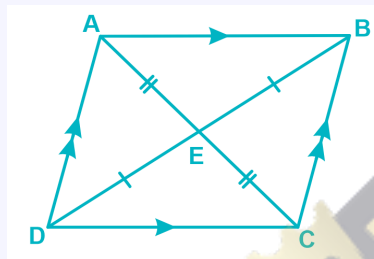
Parallelogram

- The opposite sides of a parallelogram are equal.
- The opposite angles of a parallelogram are equal.
- The diagonals of a parallelogram bisect each other.
- Diagonals may or may not be equal.
- Sum of any two adjacent angles = 180°
- Bisectors of the four angles enclose a rectangle. AS, BP, CQ and DR are angle bisectors therefore, PQRS is a rectangle.

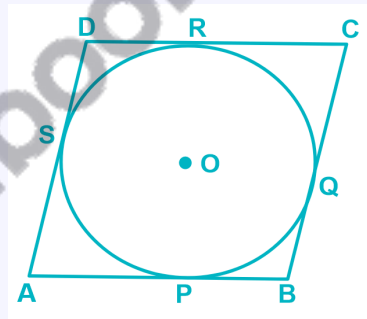


- Each diagonal divides the parallelogram into two triangles of equal area.

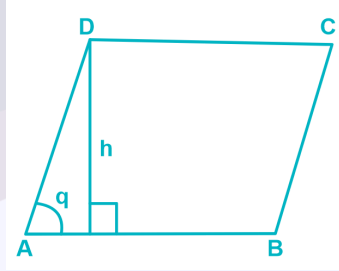
Area of $\triangle ABD$ = Area of $\triangle BDC$



- A parallelogram inscribed in a circle is always a rectangle.
- A parallelogram circumscribed about a circle is always a rhombus.



- Area of Parallelogram :
 $\text{Area} = \text{Base} \times \text{Height}$
 $\text{Area} = AB \times AD \sin \theta$



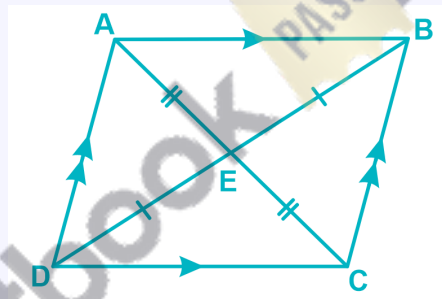
→ Area = $\frac{1}{2} \times \text{diagonal} \times (\text{sum of perpendicular dropped on it})$

→ Area = $2 \sqrt{s(s-a)(s-b)(s-d)}$

Where, $s = (a + b + d) / 2$ { d = length of diagonal }

- In a parallelogram the sum of the square of the diagonals = 2 × (sum of the squares of the two adjacent sides)

i.e. $AC^2 + BD^2 = 2(AB^2 + AD^2)$



Rhombus

- All four sides of the rhombus are equal in length.
- Opposite sides of a rhombus are parallel to each other.
- Diagonals bisect each other at right angle, but they are not necessarily equal.
- Diagonals bisect the vertex angles.
- The diagonals divide the rhombus into four right-angled triangles.
- The two opposite pairs of angles within the rhombus are equal to each other.

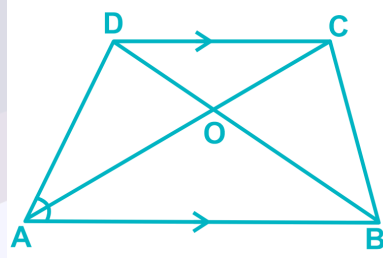
- Sum of any two adjacent angles is 180° i.e. $\angle A + \angle B = 180^\circ$
- The sum of all interior angles of a rhombus is always 360° .
- Figure formed by joining the mid-points of the adjacent sides of a rhombus is a rectangle.
- A rhombus has two lines of symmetry (the diagonals), and it has rotational symmetry of order 2 (180-degree rotation).
- Area of Rhombus = $\frac{1}{2} \times \text{Product of diagonals}$

$$= \frac{1}{2} \times AC \times BD = \frac{1}{2} \times d_1 \times d_2$$

- $d_1^2 + d_2^2 = 4a^2$
- Height of rhombus = $(d_1 \cdot d_2) / \sqrt{d_1^2 + d_2^2}$

Trapezium

- A trapezium has one pair of opposite sides that are parallel. These are called the bases of the trapezium. The other pair of sides, which are not parallel, are called the legs (or lateral sides).
- The non-parallel sides can be of different lengths and are not parallel to each other.
- The consecutive interior angles between a base and a leg are supplementary, meaning they add up to 180 degrees.
- The height (or altitude) of a trapezium is the perpendicular distance between the two bases.
- In a general trapezium, the diagonals intersect at some point and can have different lengths.
- Diagonals intersect each other proportionally in the ratio of lengths of parallel sides. i.e. $\frac{AO}{CO} = \frac{DO}{OB} = \frac{AB}{CD}$



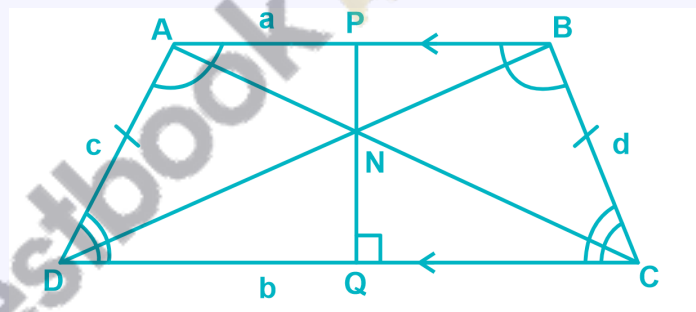
→ Area of $\triangle AOD$ = Area of $\triangle BOC$

→ $AC^2 + BD^2 = AD^2 + BC^2 + 2(AB \times CD)$

- Area of trapezium = $\frac{1}{2} \times (\text{sum of parallel sides} \times \text{height})$

$$= \frac{1}{2} \times (AB + CD) \times h$$

Isosceles Trapezium : An isosceles trapezium is a trapezium in which the non-parallel sides are equal in measure.



- $AD = BC$
- $\angle DAB = \angle CBA$
- Diagonals are equal, $AC = BD$
- If a trapezium is inscribed in a circle, it must be one isosceles trapezium.
- $\angle ADC + \angle DAB = \angle ABC + \angle BCD = 180^\circ$
- Area of Isosceles Trapezium = $h/2 (a+b)$
- The length in terms of a and b, of a parallel line segment (i.e. EF) through the intersection of diagonals of the isosceles trapezium is $= \frac{2ab}{a+b}$

